Radiology: Artificial Intelligence

DATA RESOURCES

A Pipeline for Automated Quality Control of Chest Radiographs

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Author affiliations, funding, and conflicts of interest are listed at the end of this article.

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See also commentary by Yanagawa and Sato in this issue.

Supplemental material is available for this article.

Radiology: Artificial Intelligence 2025; 7(3):e240003 • https://doi.org/10.1148/ryai.240003 • Content codes: AI CH • Published under a CC BY 4.0 license.

Despite the potential of artificial intelligence to help analyze medical imaging, a systematic review performed 1 year after the onset of COVID-19 revealed a lack of clinically viable radiologic models (1). One major reason for this is shortcut learning, in which models learn unintended, nonpathologic predictors, such as patient positioning or text annotations, which are spuriously correlated with the outcome (Fig S1) (2–6). This phenomenon is not exclusive to COVID-19 (7) or certain model architectures, and adding additional data does not necessarily help (8). Collecting more accurately annotated, less biased data can reduce shortcut learning, enhancing the generalization of downstream artificial intelligence models (2). Comprehensive data analysis and quality control (QC) before model development are needed to identify potential issues, allowing developers to mitigate their impact through techniques such as stratification and adversarial training (2,8,9).

Using large, radiologist-annotated multinational datasets, we developed a comprehensive, freely available pipeline of classic statistical and artificial intelligence–based tools for automated QC and standardization of chest radiographs called AutoQC. This pipeline aims to automatically curate large datasets to give developers an overview and statistics, while suggesting a subset of data for targeted manual review. To ensure the accuracy of our pipeline across a wide range of radiographic equipment and datasets, we assess the performance of our tools on external datasets as compared with expert annotations.

Materials and Methods

The Brent Research Ethics Committee provided ethical approval for this retrospective study (Integrated Research Application System: 282705, Research Ethics Committee: 20/Health Research Authority/2504, R&D: A095585). Informed consent was not required.

Datasets

We used two large multicenter datasets from the United Kingdom and the Valencia region of Spain for development, with three other independent datasets reserved for model testing (Table 1). A single radiograph was randomly selected for any patient with multiple images to minimize patient-level bias.

The training data were partitioned into training (64%), tuning (also known as *validation*, 16%), and test sets (20%). We used stratification to ensure consistent label proportions across partitions, which included known shortcuts and biases identified from the current literature: equipment manufacturer (2,7,10), deviation index (as a surrogate for exposure and contrast) (8,11), sex and gender (2,4,12,13), age (4,12–14), and ethnicity (12,15–17). When these data were missing, the patients were excluded. If an age range was provided, the midpoint was used.

Data Annotation

An interdisciplinary team of radiologists and data scientists identified potentially spurious correlations for inclusion in the AutoQC pipeline (Fig 1). Given the unambiguous nature of the task, radiographs were labeled for training and testing by a single radiologist in training (I.A.S., 4 years of chest radiography experience), who was supervised by an experienced cardiothoracic radiologist (J.B., 15 years of chest radiography experience). We used a custom graphic user interface to facilitate annotation *(https://gitlab.developers.cam.ac.uk/maths/cia/covid-19-projects/speedyannotate)* (Fig 2A).

The QC Pipeline

AutoQC is organized into four stages, each standardizing an image ready for the next stage and marking undesirable radiographs for review or exclusion depending on the settings (Table 2; Figs 1, S2). Table 2 provides the exclusion criteria for the development at each stage. Appendixes S3–S6 give detailed methods for each tool, organized by stage in the pipeline.

After a user runs the pipeline, an interactive report is generated, providing statistics and graphic visualizations of the QC labels and the image metadata (Fig 2B). The AutoQC Python package and all code used are available on GitLab *(https://gitlab.developers.cam.ac.uk/maths/cia/covid-19-projects/autoqc)* with documentation *(https://maths.uniofcam.dev/cia/covid-19-projects/autoqc/)*. The multiplatform package can be run on a central processing unit, multiple central processing units, a graphics processing unit, or a server. To calculate runtimes, 250 radiographs were randomly selected from each test set and run through the pipeline.

Model Evaluation

All tools were evaluated using the following metrics, calculated using bootstrapping of the test set (n = 1000): area under the receiver operating characteristic curve, accuracy, precision (also known as *positive predictive value*), recall (also known as *sensitivity* or *true-positive rate*), specificity, negative predictive value, and F1 score. The scikit-learn Python library (18) was used to sample the data and calculate the metrics.

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Abbreviation

QC = quality control

Summary

This article presents a suite of quality control tools for chest radiographs based on traditional and artificial intelligence methods, developed and tested with data from 39 centers in seven countries.

Key Points

- An open-source, publicly available automated quality control tool was developed that can reproducibly and quickly curate large datasets of chest radiographs, allowing radiologists to concentrate on more complex development tasks.
- The tool enhances the efficiency of expert quality control by providing standardized data extraction and preprocessing, statistical insights and visual summaries, and intelligently recommended subsets of data for targeted manual expert review or exclusion.
- Two large multicenter datasets from the United Kingdom and Spain were used for development, with three other independent datasets reserved for model evaluation to ensure robust assessment of generalization across diverse patient populations, imaging equipment, and clinical settings.

Keywords

Artificial Intelligence, Automated Data Curation, Automated Quality Control, Thoracic Radiography, Data Standardization

Evaluation of the Impact of Annotation and Text Removal on Classification of Radiographic Projection

To demonstrate the pipeline's beneficial impact, we compared the results of frontal projection models trained on images with text removed using our tool, the original radiographs, and radiographs with text obscured using bounding boxes of zero intensity. Each model was evaluated on data with and without text removed and on a subset with no annotations either intrinsically or removed by manual cropping. To quantitatively analyze which model focused on areas of annotations the least, we calculated the mean intersection over union (also known as the Jaccard index) for the annotation masks and the gradient-weighted class activation maps for each model.

Results

Dataset Characteristics

Data from five repositories covering 39 centers across seven countries were used, including 65 149 radiographs from 22 408 patients. After applying inclusion criteria and randomly selecting one image for each patient, 21 862 radiographs were included in the study.

Model Performance

The metrics for evaluating the tools on external test sets are summarized in Table 3, with the corresponding confusion matrices presented in Figure S7. The most crucial results are captured in Table 3, and detailed analyses for each tool are provided in Appendixes S3–S6. Additional significant findings not featured in Table 3 are summarized as follows:

Pixel intensity transformations.

The coefficient of variation, mean SD, and mean entropy of the radiographs all decreased after look-up table application. The standardized mean difference was reduced for all metrics between different manufacturers (Appendix S3).

Annotation and text removal.

The tool entirely removed annotations from 88.5% (95% CI: 86.6, 90.4), with 8.9% (95% CI: 7.2, 10.6) having none to remove. The tool partially removed annotations for 2.3% (95% CI: 1.4, 3.2) and failed to remove any for 0.4% (95% CI: 0, 0.8). A total of 0.47% (95% CI: 0.06, 0.89) experienced inappropriate inpainting within the lungs due to jewelry or metallic clothing artifacts (Appendix S4).

Cropping.

Most radiographs underwent cropping (57.2%; 4319 of 7556; Fig S34), with no inappropriate cropping identified (Appendix S4).

Aspect ratio.

The optimal bounds were 0.85 and 1.48 for the development set (Appendix S4).

Quality scoring.

The sensitivity and specificity for a poor-quality image were 34.0% (95% CI: 28.8, 39.6) and 96.7% (95% CI: 93.9, 98.2), respectively. The low sensitivity resulted from a failure to identify imperfections such as scratches or pixel drop-out, particularly prevalent in the RSNA International COVID-19 Open Radiology Database Release 1c dataset. When images with these artifacts were excluded, the sensitivity increased to 100% (95% CI: 97.0, 100.0), successfully identifying radiographs with excessively high sharpness, low contrast, blurring, and large collimation or residual padding areas.

Inconsistencies in Digital Imaging and Communications in Medicine Metadata

While evaluating the tools, we found Digital Imaging and Communications in Medicine metadata unreliable in several areas. For example, the Series Description header contained a projection in only 56.5% (12 580 of 22 274) and would have incorrectly labeled a lateral radiograph as a frontal projection in 2.9% (364 of 12 580).

Additional Results and Runtimes

Additional results tables and charts are also provided: the patient demographics within the partitions (Table S1), flowchart of studies (Fig S6), model and tuner hyperparameters (Table S4), model thresholds (Table S5), results by external test set (Tables S6–S15), receiver operating characteristic and calibration curves with confusion matrices for each test set (Figs S9–S16), and examples of gradient-weighted class activation maps (Figs S17–S21) and incorrect classifications (Figs S22–S28). Figures S44–S54 contain the image quality assessment outputs.

The final pipeline took approximately 17 seconds per image to process Digital Imaging and Communications in Medicine–format radiographs using 100 central processing units and 1 graphics processing unit (runtime results are in Table S3).

Effect of Text and Annotation Removal on Projection Classification

When comparing models trained to detect frontal projection with and without text removal, the model trained using the orig-

Table 1: Demographics, Inclusion Criteria, and Partition of the Datasets

Dataset	Partition	Centers	Inclusion Criteria	Date Range	No. of Radiographs (No. of Patients)	No. of Radiographs of Male Patients	No. of Radiographs of Female Patients	Patient Age (y)
National COVID-19 Chest Imaging Database	Development	22 in the UK	-Age > 18 years -Chest radiograph with a positive COVID-19 RT-PCR test, either positive or negative -Patient underwent chest imaging in the 3 weeks before or after the test -Previous imaging may be collected for patients who test positive for COVID-19	June 9, 2008–December 11, 2020	19 812 (5330)	12 148 (61.3)	7664 (38.7)	67.6 ± 17.0 (57–81)
Medical Imaging Databank of the Valencia Region COVID-19 dataset	Development	11 in Spain	All consecutive studies of patients with at least one PCR or immunologic test (IgM, IgG, or IgA) for SARS-Cov-2	December 21, 2019–June 8, 2020	25 447 (9072)	13 775 (54.1)	11 692 (45.9)	61.6 ± 18.2 (50–75)
BrixIA COVID-19	Test	One in Italy	-Age > 18 years -Chest radiograph with a positive COVID-19 RT-PCR test	March 4, 2020–April 9, 2020	4689 (2347)	3267 (69.7)	1422 (30.3)	64.4 ± 13.6 (57–72)
Cambridge University Hospital	Test	One in the UK	-Age > 18 years -Chest radiograph and RT-PCR assay test, either positive or negative	March 1, 2020–July 3, 2020	13 945 (5298)	7724 (55.4)	6221 (44.6)	65.3 ± 19.0 (53–80)
RSNA International COVID-19 Open Radiology Database Release 1c	Test	Four in the United States, Brazil, Canada, and Turkey	-Age > 18 years -Chest radiograph for suspected COVID-19 infection	All 2020 (pseudonymized)	1256 (361)	822 (65.4)	434 (34.6)	57.6 ± 15.9 (47–69)
Total	NA	39 centers in seven countries	NA	June 9, 2008–December 11, 2020	65 149 (22 408)	37 716 (57.9)	27 443 (42.1)	64.3 ± 17.9 (53–78)

Note.—Data are presented as numbers with percentages in parentheses or means ± SDs with IQRs in parentheses. The National COVID-19 Chest Imaging Database and Medical Imaging Databank of the Valencia Region have been updated to include more data since the data were initially curated and labeled. NA = not applicable, PCR = polymerase chain reaction, RT-PCR = reverse transcription polymerase chain reaction, UK = United Kingdom.

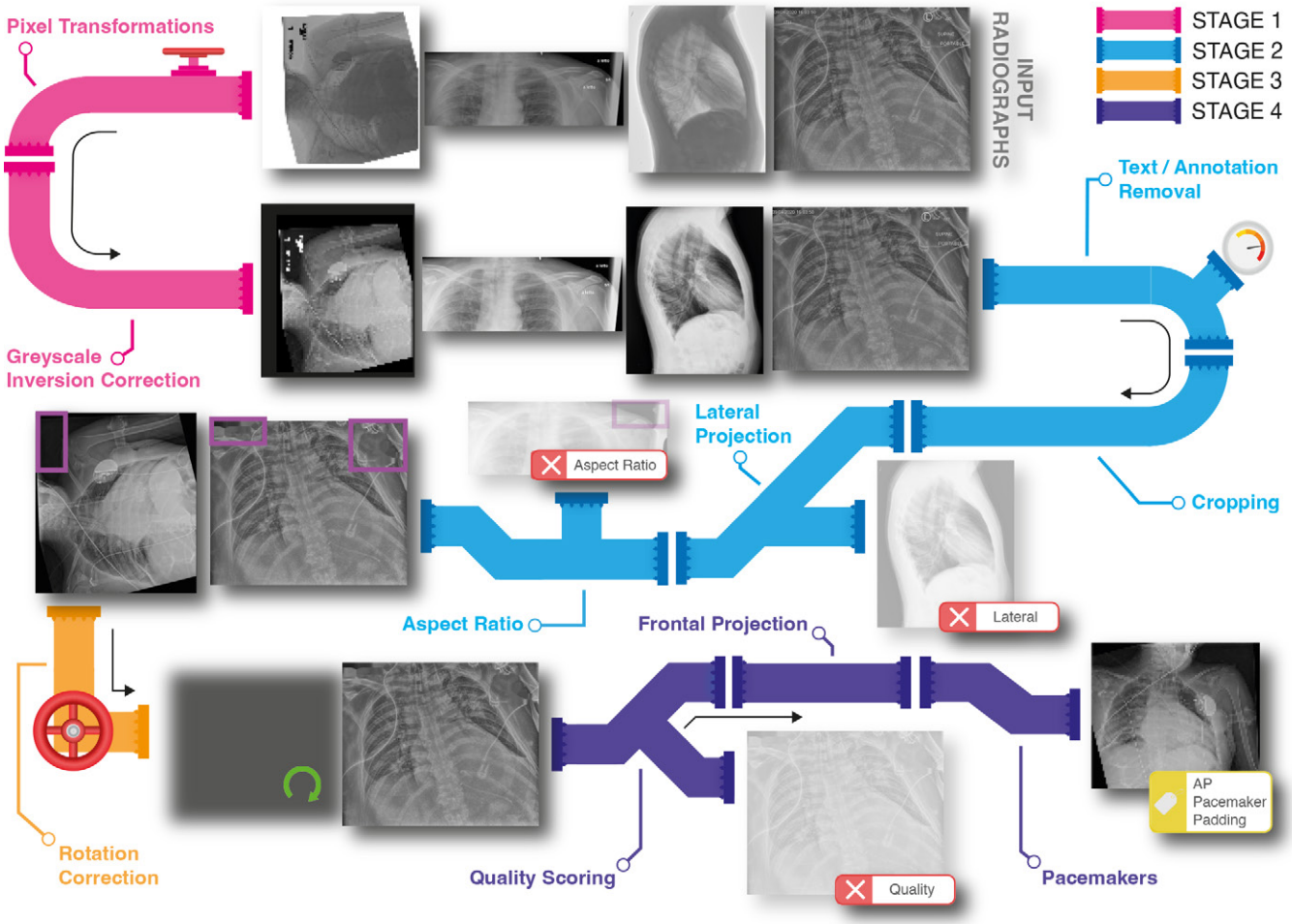

Figure 1: Graphic of the automated quality control (AutoQC) pipeline workflow with the four input radiographs at the top. Annotation removal is highlighted with purple boxes, with a green arrow showing an image after rotation correction. These four radiographs were selected for illustrative purposes, and the proportion rejected does not reflect the pipeline when deployed. AP = anteroposterior.

inal radiographs (without text removed) showed a significant decrease in prediction accuracy, sensitivity, and F1 score when tested on radiographs where the text had been removed. However, models trained using images without text performed well regardless of whether text removal techniques such as inpainting or bounding boxes were used. There was a statistically significant difference in the mean intersection-over-union scores, with the lowest indicating the least reliance on text removal and inpainting to identify projection when annotations were removed using inpainting (Table S17). The complete results of testing the deep learning projection models trained with and without text removal are shown in Figure S8 and Table S16, and a more comprehensive write-up is provided in Appendix S7.

Discussion

Our generalizable, automated pipeline can automatically curate, standardize, and provide a statistical overview of large chest radiograph datasets. It can highlight and address potential shortcuts in chest radiographs, suggesting a subset of images for a targeted review.

Radiologists must remain integral to the QC process, but improving its efficiency allows them to concentrate on more complex tasks. Furthermore, with Digital Imaging and Communications in Medicine data often incomplete or inaccurate, developers can improve performance of their models with confidence that spurious correlations have been minimized. The straightforward tool can be used with the out-of-the-box, fully trained models provided. It is freely available online to improve reproducibility, explainability, and fairness in radiographic data science. A full discussion of each tool is included in Appendixes S3–S6.

This study had limitations. The deep learning models may be susceptible to confounding, despite efforts to avoid this. The data were collected during the COVID-19 pandemic, which may reduce performance when applied to the current population. To ensure generalizability, testing on additional external datasets, such as MIMIC-CXR (19) or Chest X-ray 14 (20), is recommended, with tools being fine-tuned if performance declines. The study's limitations are discussed in detail in Appendix S8.

Additional work to further demonstrate the benefit of such automated QC tools when deployed is planned, along with continued efforts to reduce shortcut learning in artificial intelligence for chest radiographs.

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Figure 2: Screenshots of the SpeedyAnnotate annotation tool (**A**, pink) and the automated quality control (AutoQC) data explorer (**B**, blue). **(A)** SpeedyAnnotate is freely available online *(https://gitlab.developers.cam.ac.uk/maths/cia/covid-19-projects/speedyannotate)* and can be customized to use checkboxes and/or radio buttons, with the option to draw bounding boxes. The graphical user interface is built using the PyQt6 package. **(B)** The analysis package, built with Dash *(https://dash.plotly.com/)*, produces a report using the ydata profiling *(https://docs.profiling.ydata.ai)* of relevant metadata (top) and the quality control labels but also allows for interactive plotting of variables and comparison of the AutoQC outputs with Digital Imaging and Communications in Medicine metadata (bottom).

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Received January 3, 2024; revision requested February 22; revision received January 27, 2025; accepted February 5.

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Funding: I.A.S., A.B., M.R., L.E.S., J.B., C.B.S., E.S., J.R.W.M., and AIX-COVNET supported by the EU/ European Federation of Pharmaceutical Industries and Associations Innovative Medicines Initiative 2 Joint Undertaking-DRAGON (101005122); I.A.S., L.E.S., J.H.F.R., E.S., and J.R.W.M. supported by the National Institute for Health and Care Research Cambridge Biomedical Research Centre (BRC-1215-20014); J.H.F.R is partly supported by the British Heart Foundation Centre of Research Excellence (RE/24/130011), the EPSRC, and the Wellcome Trust; M.R., J.H.F.R., and C.B.S. supported by the Engineering and Physical Sciences Research Council Cambridge Mathematics of Information in Healthcare Hub EP/T017961/1; L.E.S. supported by Cancer Research UK National Cancer Imaging Translational Accelerator (C42780/A27066); A.B. supported by Cambridge Mathematics of Information in Healthcare Hub EP/T017961/1 and Austrian Science Fund (FWF, project T-1307); and M.R. and C.B.S. supported by the Trinity Challenge BloodCounts! project. The AIX-COVNET collaboration is also grateful to Intel for financial support. C.B.S. supported by the Philip Leverhulme Prize; the Royal Society Wolfson Fellowship; the Engineering and Physical Sciences Research Council advanced career fellowship EP/V029428/1; Engineering and Physical Sciences Research Council grants EP/S026045/1 and EP/T003553/1, EP/N014588/1, and EP/T017961/1; the Wellcome Innovator Awards 215733/Z/19/Z and 221633/Z/20/Z, the European Union Horizon 2020 research and innovation program under the Marie Skodowska-Curie grant agreement no. 777826 NoMADS; the Cantab Capital Institute for the Mathematics of Information; and the Alan Turing Institute.

Please note that the content of this publication reflects the authors' views and that neither Innovative Medicines Initiative nor the European Union, European Federation of Pharmaceutical Industries and Associations, or the DRAGON consortium are responsible for any use that may be made of the information contained therein.

Acknowledgments: The authors thank the Cancer Research UK National Cancer Imaging Translational Accelerator (C42780/A27066) for the use of their data repository. We want to thank the Research Governance Clinical Informatics team at Cambridge University Hospitals National Health Services Trust for their assistance with the Research Protocol. We thank the Clinical Informatics team at Cambridge University Hospitals National Health Services Trust and the University of Cambridge for their assistance in curating the Cambridge University Hospital dataset. Additionally, we would like to thank the curators of the National COVID-19 Chest Imaging Database, Medical Imaging Databank of the Valencia Region, BrixIA, and RSNA International COVID-19 Open Radiology Database Release 1C datasets, without which this work would not have been possible.

Author contributions: Guarantors of integrity of entire study, **I.A.S., J.R.W.M.**; study concepts/study design or data acquisition or data analysis/interpretation, all authors; manuscript drafting or manuscript revision for important intellectual content, all authors; approval of final version of submitted manuscript, all authors; agrees to ensure any questions related to the work are appropriately resolved, all authors; liter-

Table 2: Summary of the Purpose, Justification, and Exclusion Criteria of the Quality Control Tools

Stage	Quality Control Tools	Type of Tool	Description	Justification	Exclusion Criteria for Training
1	Pixel standardization	Statistical	Modality and volume of interest look-up tables are applied based on the image metadata. Auto-windowing is available if metadata is not.	Allows for a degree of image standardization, given that no standardized pixel unit exists, and truncates extreme pixel values.	Pediatric cases (< 18 years) Failed preprocessing procedure or corrupt
	Grayscale inversion	Deep learning	The pipeline first uses the DICOM headers to identify and correct inversion if available. Images are then classified and corrected if necessary.	Inversion varies by dataset and equipment, offering a potentially spurious correlation. Included in DICOM metadata, but metadata may be incorrect.	Dual-energy CXRs Cambridge University Hospital CXRs in the National COVID-19 Chest Imaging Database
2	Cropping	Statistical	Removes excess padding.	As a result of the digital removal of collimation at source, the presence and amount of padding may act as shortcuts, as may the size of the image and its positioning within the padding.	NA
	Aspect ratio	Statistical	Identifies CXRs that are likely to contain less than one-third of the lungs.	Deep learning models cannot accurately make predictions on fractions of the lungs. The inclusion of such images adds noise and may be spuriously correlated to the outcome.	
	Annotation and text removal	Statistical	Removes annotations and text from the image, including laterality markers.	Shown to be a frequent source of shortcuts, acting as a proxy for AP or mobile scans and, subsequently, certain types of patients (eg, acutely unwell or frail).	
	Lateral projection	Deep learning	Identifies and excludes lateral CXRs.	Use of lateral CXRs varies by center and country, with specific indications that may be spuriously correlated to outcomes.	
3	Rotation	Deep learning	Quantifies and corrects image rotation. By default, this is for 90-degree magnitudes, but the user can correct the most likely degree of rotation from 0 to 359.	Rotation may be used as augmentation when training models; however, this is usually only for angles less than 45 degrees. More than 90-degree rotation may still incorporate bias, even if augmentation is used.	Images outside of the aspect ratio range determined in stage 2 Nonthoracic CXRs Lateral projections
4	Image quality	Statistical	Identifies outliers with undesirable contrast, penetration, or postprocessing	Poor-quality images contribute to noise. Different distributions of pixel intensities and textural values between datasets offer potential shortcuts.	Extremely oblique projections Radiographs containing deep-brain stimulation devices CXRs containing only pacemaker leads or no primary device
	Frontal projection	OCR and deep learning	Classifies the image as AP or PA using the annotation mask produced in stage 2 and a deep learning model.	PA CXRs are the reference standard but cannot be performed in critically unwell or frail patients. AP CXRs are more likely to result in suboptimal patient positioning and image quality. OCR is helpful, as the technician may enter the projection.	
	Pacemaker detection	Deep learning	Detects the presence of pacemakers and similar cardiac devices, including ICDs and CRTDs. N.B. It is not intended to detect Reveal devices.	It may act as a marker of cardiovascular disease, particularly important in models for diseases with which this is a marker for poor outcomes. Additionally, this allows the feasibility of developing tools for other medical devices to be assessed.	

Note.—AP = anteroposterior, CRTD = cardiac resynchronization therapy device, CXR = chest radiograph, DICOM = Digital Imaging and Communications in Medicine, ICD = implantable cardioverter-defibrillator, NA = not applicable, N.B. = note well, OCR = optical character recognition, PA = posteroanterior.

Table 3: Results of Testing of AutoQC Tools on the Hold-out Sets

QC Tool	Accuracy (%)	Precision (%)	Recall (%)	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	F1 Score (%)
Inverted								
Internal	99.5 (99.2, 99.8)	99.2 (98.4, 99.8)	99.0 (98.2, 99.7)	99.0 (98.2, 99.7)	99.7 (99.3, 99.9)	99.2 (98.4, 99.8)	99.6 (99.3, 99.9)	99.1 (98.6, 99.6)
External	99.5 (99.3, 99.7)	99.5 (99.3, 99.7)	99.4 (99.1, 99.6)	99.4 (99.1, 99.6)	99.6 (99.4, 99.8)	99.5 (99.3, 99.7)	99.5 (99.2, 99.7)	99.4 (99.3, 99.6)
Lateral								
Internal	98.8 (98.3, 99.2)	94.6 (92.2, 96.6)	97.9 (96.3, 99.2)	97.9 (96.3, 99.2)	98.9 (98.4, 99.4)	94.6 (92.2, 96.6)	99.6 (99.3, 99.8)	96.2 (94.8, 97.5)
External	99.1 (98.9, 99.3)	34.1 (25.0, 44.4)	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	99.1 (98.9, 99.3)	34.1 (25.0, 44.4)	100.0 (100.0, 100.0)	50.6 (40.0, 61.4)
Aspect ratio								
Internal	97.3 (96.4, 98.1)	5.0 (0.0, 12.5)	25.1 (0.0, 60.0)	25.1 (0.0, 60.0)	97.6 (96.9, 98.4)	5.0 (0.0, 12.5)	99.6 (99.3, 99.9)	8.2 (0.0, 20.0)
External	97.0 (96.6, 97.4)	14.3 (9.8, 19.4)	43.6 (32.8, 55.8)	43.6 (32.8, 55.8)	97.5 (97.2, 97.9)	14.3 (9.8, 19.4)	99.5 (99.3, 99.6)	21.5 (15.1, 28.2)
Rotation (natural orthogonal)								
Internal	99.4 (99.1, 99.8)	19.4 (6.7, 42.8)	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	99.4 (99.1, 99.8)	19.4 (6.7, 42.8)	100.0 (100.0, 100.0)	31.4 (12.5, 59.9)
External	99.7 (99.6, 99.8)	6.7 (3.1, 15.8)	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	99.7 (99.6, 99.8)	6.7 (3.1, 15.8)	100.0 (100.0, 100.0)	12.4 (6.1, 27.3)
Rotation (synthetic orthogonal)								
Internal	97.7 (97.0, 98.3)	98.8 (98.0, 99.5)	96.5 (95.3, 97.7)	96.5 (95.3, 97.7)	98.9 (98.1, 99.5)	98.8 (98.0, 99.5)	96.6 (95.4, 97.7)	97.7 (96.9, 98.3)
External	98.7 (98.4, 98.9)	99.8 (99.6, 99.9)	97.5 (97.0, 98.0)	97.5 (97.0, 98.0)	99.8 (99.6, 99.9)	99.8 (99.6, 99.9)	97.6 (97.0, 98.1)	98.6 (98.3, 98.9)
OCR projection								
Internal	95.9 (93.8, 97.9)	98.6 (97.0, 99.6)	96.5 (94.2, 98.5)	96.5 (94.2, 98.5)	93.3 (86.2, 98.4)	98.6 (97.0, 99.6)	84.7 (75.0, 92.7)	97.5 (96.1, 98.7)
External	99.7 (99.5, 99.8)	99.9 (99.8, 100.0)	99.8 (99.7, 99.9)	99.8 (99.7, 99.9)	94.7 (89.1, 98.8)	99.9 (99.8, 100.0)	89.8 (83.3, 95.3)	99.8 (99.8, 99.9)
Deep learning projection								
Internal	95.4 (94.4, 96.4)	95.5 (94.2, 96.8)	96.0 (94.7, 97.3)	96.0 (94.7, 97.3)	94.6 (93.1, 96.0)	95.5 (94.2, 96.8)	95.2 (93.7, 96.7)	95.8 (94.8, 96.7)
External	92.5 (91.9, 93.2)	99.0 (98.7, 99.2)	92.3 (91.7, 93.0)	92.3 (91.7, 93.0)	93.9 (92.5, 95.4)	99.0 (98.7, 99.2)	65.9 (63.3, 68.4)	95.5 (95.1, 95.9)
Pacemaker								
Internal	91.1 (89.7, 92.5)	50.3 (44.0, 56.9)	84.7 (78.7, 90.5)	84.7 (78.7, 90.5)	91.7 (90.1, 93.1)	50.3 (44.0, 56.9)	98.4 (97.7, 99.0)	63.0 (57.3, 68.9)
External	88.3 (87.5, 89.0)	28.3 (25.7, 30.9)	91.3 (88.4, 94.1)	91.3 (88.4, 94.1)	88.1 (87.3, 88.9)	28.3 (25.7, 30.9)	99.5 (99.3, 99.7)	43.2 (39.9, 46.3)
Quality								
Internal	92.4 (87.9, 95.4)	83.3 (77.5, 87.9)	64.5 (57.6, 70.8)	64.5 (57.6, 70.8)	97.6 (94.4, 99.0)	83.3 (77.5, 87.9)	93.7 (89.4, 96.3)	72.7 (66.1, 78.5)
External	85.3 (80.8, 88.9)	69.2 (63.7, 74.2)	34.0 (28.8, 39.6)	34.0 (28.8, 39.6)	96.7 (93.9, 98.2)	69.2 (63.7, 74.2)	86.8 (82.5, 90.2)	45.6 (40.0, 51.3)
Quality (excluding artifacts)								
Internal	96.7 (92.6, 98.6)	82.4 (75.5, 87.6)	87.5 (81.3, 91.8)	87.5 (81.3, 91.8)	97.8 (94.1, 99.2)	82.4 (75.5, 87.6)	98.5 (95.1, 99.6)	84.8 (78.3, 89.7)
External	95.2 (90.0, 97.8)	53.8 (45.2, 62.3)	100.0 (97.0, 100.0)	100.0 (97.0, 100.0)	95.0 (89.6, 97.6)	53.8 (45.2, 62.3)	100.0 (97.0, 100.0)	70.0 (61.5, 77.3)

Note.—Data in parentheses are 95% CIs. The internal dataset is the hold-out test set of the combined National COVID-19 Chest Imaging Database and Medical Imaging Databank of the Valencia Region data. The external dataset combines the BrixIA, Cambridge University Hospital, and RSNA International COVID-19 Open Radiology Database Release 1c test sets. We provide full results broken down by individual datasets in Tables S6-S16 with the total numbers of each label used. Due to a small external cohort of radiographs that were naturally rotated by an orthogonal angle, we tested on both the original data (natural orthogonal) and with the same data, but with half of the test sets rotated by a random orthogonal angle (synthetic orthogonal). NPV = negative predictive value, OCR = optical character recognition, PPV = positive predictive value, QC = quality control.

ature research, **I.A.S., E.G.S., A.B., M.R., J.B., J.H.F.R., J.R.W.M.**; clinical studies, **I.A.S., M.R., J.B., J.H.F.R., E.S.**; experimental studies, **I.A.S., A.B., M.R., L.E.S., J.R.W.M.**; statistical analysis, **I.A.S., E.G.S., A.B., M.R., J.R.W.M.**; and manuscript editing, all authors

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Data sharing: All datasets are available for academic use free of charge. The National COVID-19 Chest Imaging Database (NCCID) is available upon application at the website of the NHS AI Lab *(https://nhsx.github.io/covid-chest-imaging-database/)*. The Medical Imaging Databank of the Valencia Region COVID-19 (BIMCV-COVID-19) dataset (Iter1+2) is available at *https://bimcv.cipf.es/bimcvprojects/bimcv-covid19/* (DOI: 10.21227/w3aw-rv39). The BrixIA COVID-19 dataset is available at *https://brixia.github.io/* (DOI: https://doi.org/10.1016/j.media.2021.102046). The RSNA International COVID-19 Open Radiology Database Release 1c (RICORD-1c) is available at *https://wiki.cancerimagingarchive.net/x/CaEvB* (DOI: https://doi.org/10.7937/91ahv663). The CUH dataset is available by application to the Research and Development Team at Cambridge University Hospital (cuh.research@nhs.net). The tools and the Auto-QC python package (including code, notebooks, and trained models) are available at *https://gitlab.developers.cam.ac.uk/maths/cia/covid-19-projects/autoqc*, with accompanying documentation available at *https://maths.uniofcam.dev/cia/covid-19-projects/autoqc/*. Any other data generated by the authors or analyzed during the study are available from the corresponding author by request.

Disclosures of conflicts of interest: **I.A.S.** No relevant relationships. **E.G.S.** No relevant relationships. **A.B.** No relevant relationships. **M.R.** No relevant relationships. **L.E.S.** CRUK National Cancer Imaging Translational Accelerator (NCITA) [C42780/A27066] and National Institute of Health Research (NIHR) Cambridge Biomedical Research Centre (BRC-1215-20014 and NIHR203312). **J.B.** No relevant relationships. **J.H.F.R.** No relevant relationships. **N.A.W.** Payment made to institution, CRUK grant A24042, UK Space Agency grant SST/W002469/1, Science and Technologies Facility Council-UK Research and Innovation grant ST/T003081/1. **E.S.** Consulting fees, Bracco; honoraria, GE, Bracco; payment for expert testimony, GE, Bracco; support for attending meetings/travel, GE, Bracco; co-founder and shareholder of Lucida Medical. **C.B.S.** No relevant relationships; senior consultant to the editor for *Radiology: Artificial Intelligence*. **J.R.W.M.** Treasurer, British Society of Cardiovascular Imaging; Board of Directors, Society of Cardiovascular CT; associate editor of *Radiology: Cardiothoracic Imaging*.

References

1. Roberts M, Driggs D, Thorpe M, et al. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. Nat Mach Intell 2021;3(3):199–217.
2. DeGrave AJ, Janizek JD, Lee SI. AI for radiographic COVID-19 detection selects shortcuts over signal. Nat Mach Intell 2021;3(7):610–619.
3. Driggs D, Selby I, Roberts M, et al. Machine Learning for COVID-19 Diagnosis and Prognostication: Lessons for Amplifying the Signal While Reducing the Noise. Radiol Artif Intell 2021;3(4):e210011.
4. Jabbour S, Fouhey D, Kazerooni E, Sjoding MW, Wiens J. Deep Learning Applied to Chest X-Rays: Exploiting and Preventing Shortcuts. ArXiv 2009.10132 [preprint] . Posted September 21, 2020. Accessed April 12, 2022.
5. Garcia Santa Cruz B, Bossa MN, Sölter J, Husch AD. Public Covid-19 X-ray datasets and their impact on model bias - A systematic review of a significant problem. Med Image Anal 2021;74:102225.
6. López-Cabrera JD, Orozco-Morales R, Portal-Diaz JA, Lovelle-Enríquez O, Pérez-Díaz M. Current limitations to identify COVID-19 using artificial intelligence with chest X-ray imaging. Health Technol (Berl) 2021;11(2):411–424.
7. Zech JR, Badgeley MA, Liu M, Costa AB, Titano JJ, Oermann EK. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. PLoS Med 2018;15(11):e1002683.
8. Geirhos R, Jacobsen JH, Michaelis C, et al. Shortcut Learning in Deep Neural Networks. Nat Mach Intell 2020;2(11):665–673.
9. Zhao Q, Adeli E, Pohl KM. Training confounder-free deep learning models for medical applications. Nat Commun 2020;11(1):6010.
10. Biondetti GP, Gauriau R, Bridge CP, Lu C. KPA KP. "Name that manufacturer". Relating image acquisition bias with task complexity when training deep learning models: experiments on head CT. ArXiv 2008.08525 [preprint] https://arxiv.org/abs/2008.08525. Posted August 19, 2020. Accessed April 18, 2024.
11. Dave JK, Jones AK, Fisher R, et al. Current state of practice regarding digital radiography exposure indicators and deviation indices: Report of AAPM Imaging Physics Committee Task Group 232. Med Phys 2018;45(11):e1146–e1160.
12. Duffy G, Clarke SL, Christensen M, et al. Confounders mediate AI prediction of demographics in medical imaging. NPJ Digit Med 2022;5(1):188.
13. Glocker B, Jones C, Roschewitz M, Winzeck S. Risk of Bias in Chest Radiography Deep Learning Foundation Models. Radiol Artif Intell 2023;5(6):e230060.
14. Brown A, Tomasev N, Freyberg J, Liu Y, Karthikesalingam A, Schrouff J. Detecting shortcut learning for fair medical AI using shortcut testing. Nat Commun 2023;14(1):4314.
15. Gichoya JW, Banerjee I, Bhimireddy AR, et al. AI recognition of patient race in medical imaging: a modelling study. Lancet Digit Health 2022;4(6):e406–e414.
16. Delgado J, de Manuel A, Parra I, et al. Bias in algorithms of AI systems developed for COVID-19: A scoping review. J Bioeth Inq 2022;19(3):407–419.
17. Brown RCH, Savulescu J, Kelly D, Wilkinson D. A public health ethic should inform policies on COVID-19 immunity passports - Authors' reply. Lancet Infect Dis 2021;21(4):456–457.
18. Metrics and scoring: quantifying the quality of predictions — scikit-learn documentation (version 1.4.2). https://scikit-learn.org/stable/modules/model_evaluation.html#model-evaluation. Accessed April 19, 2024.
19. Johnson AEW, Pollard TJ, Berkowitz SJ, et al. MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports. Sci Data 2019;6(1):317.
20. Wang X, Peng Y, Lu L, Lu Z, Bagheri M, Summers RM. ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2017; 3462–3471.